

Vulnerability, Sharing, and Privacy: Analyzing Art Therapy for Older Adults with Dementia

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ABSTRACT

Older adults are most often considered consumers of online information, but recent work highlights the importance of engaging older adults in content generation and online sharing. One context in which older adults generate and share content is art therapy for individuals with dementia. Our analysis draws on Altman's notion of privacy and territorial regions to understand what sharing means for this vulnerable population. This theoretical framing reveals the ways in which privacy is cooperatively negotiated, which is in contrast to the individualistic view of existing sharing systems; how older adults derive benefits from sharing depending on interaction with their audience; and how sharing fluctuates between a focus on the process of therapy versus the product depending on privacy needs. Our analysis contributes an understanding of the complex nature of sharing for vulnerable populations and offers design considerations for systems that support this practice.

Author Keywords

Art therapy; older adults; dementia; sharing; privacy

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

INTRODUCTION

A topic of interest to CSCW is how people create and share content online, both in terms of who is producing content and how new tools can support this societal practice. Older adults are most often considered consumers of online content, but recent work has begun to focus on engaging older adults in digital content generation [27,36,38] and online sharing with peers or family members [7,33,39]. There is much to learn from how older individuals engage in content generation and sharing in primarily offline contexts. As one offline context, this paper examines the practices of older adults with dementia and their art therapists. Art therapy provides a positive social context in which older adults with dementia are empowered to create and share content and engage with

others in meaningful discussion, reflection, and self-expression [27,36]. In this way, our work challenges normative views of dementia as a disease in which the body “breaks down” and needs repair [17].

More than five million people in the U.S. have age-related dementia, which clinical research defines as a “decline in cognitive performance” that affects memory and language, and Alzheimer's disease accounts for approximately 70% of these cases [21]. By 2050, it is projected that almost 16 million people will have Alzheimer's [2]. This paper presents a year-long qualitative study of the practice of creating and sharing therapeutic artwork involving older adults with dementia. The artwork serves as a rich communication artifact throughout different contexts of sharing. Sharing in this case involves the exchange of information about oneself in addition to and as part of this artifact. Art therapy is a particularly compelling context for understanding how vulnerability, risk, ownership, and privacy are negotiated in the practice of sharing. Our analysis examines the ways in which these older adults, their therapists, and other stakeholders share artwork through the theoretical lens of Altman's territories. Altman defines privacy as the “desired access to self and others at any moment in time,” [1]. He describes how people negotiate privacy boundaries through various regions of disclosure when co-located in physical space. Our work and others in CSCW [23,40] use this framing to understand the social practices around sharing and privacy.

Altman's notion of regional territories and typology provides an analytic frame for examining art therapy, through which we identify four regions of sharing: *primary*, *secondary*, *surrogate*, and *public*. We articulate regional sharing behaviors in the context of vulnerability, including: the older adult's role in cooperative negotiations of privacy; the effects of a co-present or visible audience on personal enrichment derived from sharing; and how highlighting the process versus the product of art therapy affects privacy in sharing. This work makes two primary contributions: (1) a nuanced understanding of the nature of sharing for a particular vulnerable population, in which sharing is a social process, artifacts are co-owned, and privacy boundaries are cooperatively negotiated; and (2) characterization of specific sharing practices among older adults with dementia within art therapy, which reveals considerations for designing systems to support the complex and dynamic nature of sharing within this context.

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RELATED WORK

We highlight existing work on how older adults produce content and share online as well as tools to facilitate these activities within the context of art therapy.

Content Production by Older Adults

Older adults are often seen as consumers of online content [39]. Older adults may lurk in online communities rather than publish content due to the complexity of online interfaces for sharing content [25]. Furthermore, new technologies for older adults may not provide sufficient means for reciprocal communication involving this content [11,25]. To address this, recent work focuses on how to engage older adults in creative activities [38] and content sharing with peers or family members [7,33,39]. Yet, few systems facilitate creative production by older adults. As one example, Waycott et al. [39] developed the *Enmesh* system to facilitate creating and sharing of photos and stories through an iPad application. A field deployment of *Enmesh*, which enabled sharing of photos throughout the day, identified how art in the home is used to share a historical life narrative. Their results also suggest sharing with peers was perceived as less meaningful than it would have been to share with family members. Further, this work highlights a need to develop new technologies for older adults that provide opportunities for self-expression and creativity [39].

Prior work, however, focuses on supporting independent older adults, and does not take into account changes in older adults' abilities [15]. For example, older adults with dementia experience changes in cognitive abilities, which affect how they use technology and participate in creative activities (e.g. art therapy) [36]. Recent systems aim to support the creative process for older adults with dementia [27] but do not consider the ways in which sharing may be supported in the context of vulnerability. This prior work notes the importance of new digital tools to support creation and sharing among older adults, particularly with vulnerable older adults. The present paper investigates how older adults with dementia create and share in the context of art therapy, and one contribution is to guide the design of new technologies that support sharing for this population.

Systems for Sharing Online

Producing content can help older adults express themselves and be creative, but an important aspect of producing content is the act of sharing content. House et al. [20] describe the importance of sharing photos in creating and sustaining relationships, constructing memory, self-expression, and self-presentation. Older adults often think of sharing content with relatives, such as children and grandchildren [33], and while prior work introduces tools for sharing visual content (e.g., photos) [11,12], this sharing is often done by social contacts rather than the older adult. Similarly, Blythe et al. [6] explored prototypes to share visual content with older adults in a residential care setting but relied on professional artists to create the content. Nonetheless, these prototypes and associated content encouraged curiosity and engaged older adults in conversation. While we focus on older adults

with dementia, systems that facilitate content production and sharing for other vulnerable populations, such as children (e.g., [31]), provide additional fodder for understanding issues of vulnerability and privacy online.

Supporting Sharing in Art Therapy

Art therapy is a rich context for content generation and sharing, but relatively few systems have been developed to support this practice. Researchers have relied on feedback from art therapists to inform the design of technologies to facilitate art therapy sessions [5,15,19]. A survey of 133 art therapists [27] found that technologies should support customization and adaptability, assessment of individual needs, empowerment, and simplicity. Moreover, this study and others found that technology should not replace art therapists; instead, systems should play a supplementary role in the creation and sharing processes [5,7,27]. The few systems that have been developed for art therapy focus on supporting offline interaction, such as participation, conversation, and individualized support. The ePAD prototype helps prompt and monitor participation in art therapy [27]. Blank Canvas, a canvas in a main lounge of an assisted living community, and Video Window, a screen playing video of the environment outside of the community, prompted conversation and expression amongst residents [6]. Other systems [5,19] use computer-based algorithms to provide adaptive, automated assistance to seniors in art therapy sessions but find “engagement is very individualized” and tools should support individualized content creation. These systems identify several challenges of designing tools to support the practice of art therapy, but prior work has yet to detail considerations for developing new technologies for online content sharing within this context. Our analysis provides practical considerations for system design for this particular context but also contributes a more nuanced understanding of sharing and privacy for older adults with dementia, which challenges the ways in which current online systems support vulnerability.

METHOD

Our qualitative inquiry took place over one year, and the results in this paper are derived from: (1) participant observations, (2) focused observations of art therapy, and (3) in-depth interviews with art therapists.

Participant Observations

Our understanding of art therapy began with two authors serving as volunteers in an art therapy program for older adults with dementia within a residential facility. We volunteered for two hours a week for seven months, assisting the community's art therapist. In this capacity, we focused on our own tacit learning of the practice of art therapy through apprenticeship under the art therapist and did not collect data about older adults. By assisting the art therapist, we learned how to prepare appropriate and individualized materials for older adults with dementia making art, how to support older adults with varying abilities in creating artwork (e.g., painting with an adult hand-over-hand, writing titles or commentary onto a piece of work), and how to talk with

older adults while creating and sharing artwork. At times, we even created and shared artwork ourselves alongside residents of the community as part of a weekly group art therapy session. In addition to learning about the practice of art therapy through hands-on participation in the art studio, the researchers engaged in reflective discourse with the art therapist about the process of art therapy, tools used to support making art, and sharing artwork.

Field Observations

During the second half of this year-long project, three researchers conducted 11 focused observation sessions of art therapy within this community. We obtained human subjects approval from the community to conduct general observations of art therapy. We also obtained consent from the older adults with dementia and their authorized representative before studying their participation in art therapy in detail. Each observation session lasted 1.5-2 hours and involved both one-on-one and group therapy sessions. One-on-one sessions involved an art therapist and an older adult, whereas group sessions involved between 2-9 older adults. We observed 21 older adults (19 female), and the vast majority of older adults had dementia from Alzheimer's disease, stroke (e.g., vascular dementia), or another condition. Besides cognitive impairment, all residents used a wheelchair or walker to travel to the art studio. Some had severe hand tremors or upper-body mobility impairments that impacted making art.

Researchers sat on opposite sides of the art studio and took detailed notes about the activities, materials, and social interactions related to art therapy. Participant observations emphasized our own learning about art therapy, whereas periods of focused observations examined the details of how older adults engaged with each other and the art therapist during the process of creating and sharing artwork. In this capacity, and per our human subjects' approval, we took on the role of outside observers and researchers did not interact with informants during the therapy sessions. When possible, we audio recorded and took photos during art therapy sessions. The researchers debriefed with the art therapist following each session.

Interviews with Art Therapists

We conducted interviews with five additional trained and certified art therapists to supplement our ethnographic work within the studied community. We note interviews with domain experts often require fewer informants to reach saturation [16], which we achieved after five participants given that interview data supplement our long-term fieldwork. Interview participants had 1-7 years of experience practicing art therapy and lead art therapy programs in nursing homes, continuous care facilities, or adult day care programs. In a typical week, participants reported conducting 4-10 hours of art therapy with 3-12 older adults diagnosed with dementia. Researchers conducted interviews lasting 60 minutes (on average) in-person or via video chat. Interviews were semi-structured, and we encouraged

participants to discuss examples from art therapy sessions. We formulated the interview protocol to uncover more details about the materials used during therapy, particular goals of art therapy, strategies for sharing artwork, and benefits and challenges of sharing. All interviews were audio recorded and transcribed.

Data Analysis

Data analysis followed a constructionist grounded theory approach [9,10], and as such our process emerged through interactions at our field site, with our data, and between members of our research team, one of whom is an art therapist. Initial data included field notes, photographs, and transcripts of recorded therapy sessions and interviews. We iteratively coded data and related the codes and emerging concepts to one another. Analysis focused on constant comparison of data, seeking to further refine and explicate our emerging understanding of the studied phenomena. Through this iterative process of coding and theorizing, we began to view Altman's work [1] as a productive theoretical lens for establishing a nuanced understanding of the concepts of sharing and privacy in this context.

Our approach requires reflexivity about how we construct our actions and the position we adopt as researchers. Our work places therapeutic clinical practice at the center of analysis. Clinical therapy emphasizes the importance and positive nature of sharing in art making [27,36]. Art therapists aim to provide clients with a sense of dignity [4], and sharing is part of this: "*at times it is appropriate for the product to be shown to peers, caretakers, family or others involved in the client's life,*" [24]. Art therapy as a practice counters the social construction of dementia as a disease in which the body breaks down and needs fixing [17]. We seek to understand when, why, and how making and sharing unfolds as a way of empowering older adults with dementia.

While art therapy can be empowering, clinical practice and our analysis positions older adults with dementia as a vulnerable population. The positioning as vulnerable is also socially constructed and framed by normative ideology. That is, positioning an individual or group as vulnerable implies needing some sort of protecting or special provisions related to possible weakness or limitation. There is a tendency in HCI to portray a particular group as vulnerable to showcase or motivate a new application, but the framing of vulnerability may further disempower individuals or make existing vulnerabilities worse [32,37]. We remain conscious of this and focus on the lived experience of individuals with dementia and what vulnerability means in the context of empowering, engaging, and providing individuals with a voice (or not) through making and sharing in art therapy [14].

THE CONTEXT AND PROCESS OF ART CREATION

Our fieldwork took place within a long-term care community in the Midwestern region of the United States. Our interview participants (art therapists) work in similar environments. Art therapy, along with physical and occupational therapy, is a service provided to older adult residents at our field site. In

this paper we use the term *resident* to describe an older adult diagnosed with dementia who participates in art therapy, both those at our field site and discussed during interviews with additional art therapists. The term *resident* conveys our focus on older adults who no longer live independently and who are no longer autonomous in their daily activities.

Art therapy sessions may involve one resident and the art therapist or the art therapist working with a small group of residents. The art therapist prepares materials and activities and guides the resident through each phase of the session: introducing a topic for art making; engaging in the act of making art; and a wrap-up, which often involves sharing the artwork through discussion of and reflection on what the older adult created. We observed art making that was primarily visual (e.g., painting, drawing, needle felting), although occasionally involved audio-visual materials (e.g., photos with audio of residents singing and describing past events stitched together). Art therapists use a wide range of materials, including colored pencils, charcoal pencils, watercolors, acrylic paint, oil pastels, clay, silk, felt, and magazine clippings. Art therapists emphasized the importance of using high quality materials (e.g., heavy weight watercolor paper) and sophisticated activities rather than “childish” materials (e.g., crayons) and tasks (e.g., finger painting).

Art therapists also use digital tools in art making, although none of the residents we worked with used computers independently. The art therapist at our field site used a computer and camera to create stop-motion animations with residents, poetry writing and recording, and multi-media compilations of resident artwork, audio recordings, and music. This art therapist also used a camera and computer to animate two-dimensional and three-dimensional artwork and personal objects. In fact, some residents who did not engage with non-digital art materials could still create stop-motion animations by manipulating familiar objects.

The role of the art therapist in constructing a meaningful, safe, and accessible context for content generation is critical. Art therapists carefully tailor the tools and their use to each resident’s needs by selecting materials that are appropriate for their cognitive abilities (e.g., complexity of verbal instructions or steps required) and physical abilities (e.g., dexterity to hold a pencil or arm mobility to reach a canvas) as well as formulating art making tasks that engage older adults’ interests and emotional needs (e.g., painting about what a particular relationship means). Often art making sessions, particularly group sessions, have an open-ended structure in which the art therapist explains the concrete goal of the session (e.g., decorate a mailbox) or the theme of the day (e.g., express love) and then makes subtle adaptations to fit the needs of each person making art.

Sometimes residents are confused about the activity, “*I don’t know what I am doing*” (R5), or show frustration due to physical ability, “*I feel like I’m painting without seeing what I’m doing*” (R4), or self-perception of art making ability “*I*

feel like a flop” (R14). Art therapists carefully respond to such statements by assessing the older adult’s situation, asking further questions about what is challenging, and offering constructive prompts to the residents. For example, during one session R11 seemed frustrated because she did not know what to paint about the topic of “love”. Rather than suggesting what to paint, the art therapist prompted the resident by saying, “*What color represents love, or just what comes to your mind?*” This structure provided by the art therapist during art making scaffolds both the *process* of therapeutic art creation and the *product* yielded during a session. As such, sometimes art therapists must take an active role in making art with a resident, acting more as a co-creator of content, due to a resident’s limited ability to manipulate certain materials (e.g., combine colors to get a new color) or to finish the artwork. For example, AT3 noted, “*I will help people finish an art piece, and that’s by permission as well. ‘Can I collaborate with you to like help you finish this?’*” Although art therapists may act as co-creators of content, or even curators (e.g., assembling multiple pieces into a mural), they intentionally attribute credit to residents as a way of empowering these older adults.

Art therapists conclude art therapy sessions by encouraging residents to share their artwork, either one-on-one with the art therapist or with other residents participating in art therapy. The art therapist pays close attention to the information shared among the group to identify and document residents’ needs or current state (e.g., affect, emotions). After a sharing session, the art therapist often writes notes on the back of the artwork (e.g., quotes from the resident) or on separate forms that are part of the older adult’s care plan, which is formal documentation of their therapy goals and progress. Therapeutic artwork is co-owned in that the resident is the primary artwork creator but the artifacts are collected, stored, and annotated by the therapist as part of the resident’s portfolio.

REGIONS OF SHARING

Our analysis of art therapy details a rich context for content generation among a vulnerable and offline population, yet a core component of the therapeutic process is the act of sharing. We draw on Altman’s typology of territories (or regions) as a lens for understanding how information sharing occurs and privacy is managed. His initial typology includes three regions: *primary*, a person or group controls their privacy; *secondary*, there is some level of control; and *public*, control is relinquished [1]. Characteristic of Altman’s early work, which examines privacy in co-located interaction, and recent analyses that apply his typology to online interaction [23,40], the boundaries of these regions are characterized by the extent to which an individual or group has control over sharing information and the gradations of private versus public sharing. Building on this work, Petronio [29] suggests collectively held privacy boundaries exist because information is not solely about the self and that individuals’ privacy boundaries change over the lifespan. In Petronio’s work [30], which extends Altman’s framing,



AT: Where would you like to be if you could be anywhere in the world?
 R: In a space-area with family... In the mountains, I would share that with my family.
 ...
 AT: If you were sending a postcard from [town], what would be in the picture?
 R: I would want it to have everyone in the family... That would be enjoyable in sharing.
 ...
 AT: Did your whole family fit into the house in [town] when they all got together?
 R: Oh yeah, very large. everybody got things, ball games and stuff like that.
 ...
 AT: Ok, if you dictate I'll write. So dear... so who's somebody in your family that you would like to send this to?
 R: Dear mom.
 AT: Ok and you would say? I wish we
 R: Had more time to have more fun with all of you.
 AT: Ok, anything else?
 R: It's been such a wonderful trip we've had with as many of us as we can.



Figure 1. Left: art therapist helps resident glue fabric to a postcard to create a mountain scene. **Middle:** transcript from the therapy session. **Right:** back of finished postcard with the resident's message.

vulnerability is inherent in how caregivers and older adults grow interdependent and take on more work of coordinating the way an older adult's privacy is controlled and access to information is regulated. Altman's typology and the widely-used notion of regions in privacy research (e.g., [23,40]) helps understand the dynamics of sharing, decision-making, and the negotiation of privacy within this particular context.

Primary Region: One-on-One with Art Therapist

The most common way in which a resident engages in sharing is through one-on-one interaction with the art therapist. This interaction is characteristic of the *primary region* in which the resident plays a central role in the act of sharing and involves a private context. Sharing in this region is an ongoing and fluid process negotiated by the resident and art therapist, but the resident plays an active role in deciding whether, how, and when they share. Often sharing occurs organically through the act of an art therapist and resident creating artwork side-by-side or co-creating an art piece. The discourse around the artwork is intimate at times, drawing out a resident's complex emotions and feelings, and achieved through rapport that exists between the therapist and resident. Sharing may involve in-the-moment assessment of how the resident is feeling, reflection on why they are making art (e.g., what is the intention behind making art), and summarizing a piece with a title or statement about what the artifact represents. In our fieldwork, we only audio recorded interactions during one-on-one therapy sessions to not disrupt the dynamic between the resident and art therapist, again emphasizing the intimacy of this context.

The art therapist may prompt a resident to identify a social contact with whom they would like to share their artwork, regardless of whether or not they will ultimately share with this person. The act of creating with the intention to share with a particular person is therapeutic in that it may bring up other social and emotional issues for the therapist to address. For example, one session involved making postcards by gluing fabric onto cardstock (see Figure 1). The discussion

during art making focused on remembering people (e.g., the resident's family and deceased mom) and a place they visited often. After completing the postcard, the therapist turned it over and asked the resident to dictate a message to her mom, which the therapist transcribed onto the back of the postcard. This act of hypothetical sharing is intimate and involves the exchange of personal information during in-the-moment making and focused creation of messages for loved ones. While we observed multiple residents wanting to share with loved ones in this region, there are many challenges (e.g., longing for a deceased contact, coordinating with relatives).

Secondary Region: Group Therapy Sessions

Another way in which sharing occurs involves group art therapy sessions, which we characterize as a *secondary region*. Here, the resident plays an active role in the sharing process deciding what information to share with a private but changing social group of peers. In this region, residents decide in-the-moment whether and how to share. Sharing involves face-to-face social interactions with art therapy participants, including other residents and volunteers. The art therapist acts as facilitator to encourage each resident to talk about their artwork with the group at the end of the session. The therapist often goes from resident to resident, first asking whether the individual would like to share their work, and if so, to tell the group about their piece. Art therapists use general prompts to encourage the resident to share a description of the artwork, such as "*What would you title this if you could title it and why?*" (AT3). These prompts encourage residents to reflect on what they created and to share with their peers a more detailed description of their artwork, often drawing out emotions or reminiscing about past events. For example, R1 painted a sunset during an art therapy group session and described what the sunset looked like when he was a pilot: "*If the clouds are below you, you could see the red [over the clouds]. It's glorious.*" In addition to these prompts, art therapists may use knowledge of each resident to facilitate sharing, such as an example

provided by AT1: *"John, you and your wife like to go to Florida every summer, do you want to talk about that?"* The therapist draws on her established relationship with the resident and intimacy characteristic of the primary region to facilitate sharing in a slightly more public region.

Sharing in group therapy facilitates social interaction among group members, often resulting in residents complementing and commenting on each other's artwork. Although receiving positive comments about the visual appeal of one's artwork is valuable, during group sessions the art therapist often refocuses discussion on the meaning and emotional significance of the artwork rather than purely aesthetic qualities. That is, sharing in this region involves disclosure of personal information, feelings, and emotions rather than simply presentation of the final artifact.

Surrogate Region: Sharing on Behalf of Others

Our analysis revealed a region that has yet to be accounted for in Altman's initial typology or by other studies of territory in online systems [23,40]. We call this the *surrogate region*, which involves a form of private or confidential sharing of artwork *on behalf of* the resident, but the resident does not typically play an active role in deciding how or when to share. Art therapists facilitate this sharing and involve individuals who have legal responsibility related to the resident's wellbeing (i.e., per HIPAA regulations [18]). In this region, art therapists act as a proxy for sharing [28].

As an example, care plan meetings are structured interactions that allow the clinical team (i.e., art therapist, occupational therapist, physical therapist, social worker, nursing assistants, nurses, doctors) to cooperatively review the resident's current status, progress in therapy, and set goals for future therapies or treatments. The resident's legal authorized representative, often a relative (e.g., spouse or child) who acts on behalf of the resident's health and financial matters, may attend these meetings. Care plan meetings occur roughly every three months, and the resident may or may not be present at the meeting. At times, additional care plan meetings are held between only the art therapist and the resident's authorized representative. During these meetings, the art therapist uses the resident's artwork to explain their interactions during art therapy sessions. The focus here is on clinical assessment, and the artwork serves as an artifact that enables the therapist to provide details into the resident's abilities, feelings, and emotions evidenced by a particular piece. Art therapists perform collective self-presentation [26] as a proxy for the resident by selecting pieces of artwork that best represent the resident's abilities and highlight the benefits of art therapy. This sharing often aims to help relatives understand the resident's abilities:

"Look at this [referring to the artwork] that your family member can still do. You may have difficulty communicating verbally with him but they can really express themselves through the artwork," (AT2).

With this information, the authorized representative may make decisions about the resident's care or give approval to

share a particular piece with a broader audience (e.g., in the *public region*, described below). Nevertheless, care plan meetings focus on clinical assessment and infrequently involve the resident's authorized representative.

Another interaction within the *surrogate region* involves art therapists engaging in informal sharing with the resident's relatives or close social contacts. Art therapists perform important relationship maintenance work on behalf of the resident by sharing aspects of artwork that may strengthen the relationship between the resident and his or her relative. For example, AT1 noted that she will share pieces *"if there is emotional significance to the person who created it,"* and provided an example: *"One of the participants had written his wife's name. So, I know that was very sweet, so after that I showed that 'hey look at this' to the wife..."* Art therapists indicated that residents' social contacts are largely unaware of sharing in art therapy but suggested that new online tools could provide visibility into this practice.

Public Region: Exhibits and Dissemination

The fourth way in which sharing occurs is through community-based art exhibits and dissemination through newspapers, newsletters, and online groups. In this region, residents do not play an active role in deciding how or with whom they share their artwork nor can they control the public or open nature of the audience viewing their work.

Art therapists report that public art exhibits occur between one and four times per year, and during these exhibits they frame and display the artwork throughout a residential community or similar public venue. In contrast with other regions, residents play a lesser role in sharing their artwork during public exhibits. Instead, art therapists plan, curate, and execute art exhibitions. AT3 explained, *"I take over the logistics, get the consent forms signed, hanging the exhibits...So, really handling all the logistics."* Prior to displaying the artwork, the art therapist explains the purpose of the exhibit to the resident and requests verbal consent to include her or his artwork in the exhibition. Additionally, each resident's authorized representative must provide written consent for public displays.

After obtaining formal approval for sharing a piece, the art therapist curates a collection for the exhibition, and sharing in the public region focuses primarily on the artifact. For example, art therapists report that aesthetics are important in selecting artwork for the exhibitions. AT2 said, *"[I] choose pieces that are kind of more aesthetically pleasing."* AT1 explained that for public exhibits she selects *"...something that is visually interesting or visually speaks [to] the skills [of] the person,"* Art therapists use public sharing via art shows to communicate the residents' artistic abilities to a broader audience (e.g., friends, staff, general public, and dementia groups). While valuable, art exhibits are infrequent due to the work of coordinating and planning these events as well as layers of approval required before a piece may be displayed publicly.

As other examples of public sharing, care facilities create paper newsletters about art therapy, and one art therapist set up a Facebook group for residents' social contacts to view artwork. Sharing artwork through newsletters and a Facebook group abides by the same consent process as public exhibitions. These forums provide social contacts with awareness of what happens in art therapy. AT3 said:

"[On Facebook] we document people when they are making the artwork, and sort of the process of that, and it's mostly geared towards families... so they can kind of see what that person is doing throughout the day."

The art therapist at our field site has considered the benefits of documenting and sharing some of what happens in art therapy through an online blog, but has not done so because of questions of privacy and ownership when working with individuals with cognitive impairment as well as the complex multi-layered approval process involving residents, their authorized representatives, and the broader organization supporting the art therapy program. AT3 stressed the potential of using social media as a channel for sharing, although she also has privacy concerns:

"I'm much more cautious in terms of things that are presented to the general public, making sure the resident understands. You know when something's out there, on the Internet, it's there forever."

Additionally, we learned that care facilities have concerns around the perceived public and uncontrollable nature of social media and are reticent to support online content sharing. We suspect that this is largely because systems have yet to address the complex and dynamic nature of privacy in this context as well as regulations (e.g., HIPAA).

REGIONAL SHARING BEHAVIORS

The concept of regions helps characterize the social practices around sharing in the context of vulnerability, which include: (1) the cooperative negotiation of privacy in sharing; (2) the effects of a co-present or visible audience on personal enrichment derived from sharing; and (3) how highlighting the process versus the product of therapy affect privacy.

Sharing and Cooperative Negotiation of Privacy

Privacy for Altman is a selective process in which individuals and groups regulate how open or closed they are while interacting with others [1]. Our analysis reveals that sharing in this context is not individualistic but rather a social process that involves cooperative decision-making about privacy. This can be viewed as a negotiation between how various actors perceive the risks of publicly disclosing information about a vulnerable individual and the ability of the resident to understand these risks and independently consent to such sharing. Hence, the positioning of this population as vulnerable implicates multiple layers of approval or consent for public sharing in order to protect the resident from potential risks [30]. Yet, this cooperative negotiation of privacy indicates possible gradations of vulnerability, in which consent to share shifts from implicit (e.g., head nods) to explicit (e.g., signed consent) agreement

as one moves towards public sharing, which is characteristic of Altman's early observations [1].

In the primary and secondary regions, residents play an active role in deciding whether and how to share. Residents decide what to share in the moment, including personal information (e.g., memories or emotions) reflected in the artwork. Art therapists verbally gauge each residents' willingness to share with the group but also watch the resident's body language and facial expressions to assess their willingness to share, attending to implicit cues. In these regions, residents dynamically decide what information they will disclose, ranging from a basic description of artwork to personal circumstances motivating the piece. However, art therapists work with each resident to encourage information sharing. AT3 explained, *"Making art that's like expressing awful things and that is maybe something that you don't wanna share,"* but then noted the benefits of receiving support while sharing negative emotions, *"I would encourage them to share it [negative emotions] and receive group feedback or support."* In this respect, the art therapist and resident negotiate what and how to share in the primary and secondary regions, which are spaces bound by confidentiality, intimacy, and social structures that encourage and enable the active role of the resident.

In contrast, residents have a limited role in sharing in the surrogate and public regions. In the surrogate region, the resident may have no say in sharing, as their authorized representative often consents on their behalf. In other situations, the resident may have some say over which pieces of artwork others may access (e.g., art therapist may verbally ask a resident for permission) but have no control over the information that may accompany the artifact (e.g., emotional state). Residents do not have mechanisms to regulate how open or closed they want to be about sharing in the surrogate region. Privacy is regulated at a high-level by the policies around care plan meetings (e.g., regulations around who may access this content) but in practice is managed by the art therapist, who chooses what and how to share content created by and information about the resident. We discuss this tension between policy and practice below.

In the public region, residents play a minimal, if any, role in sharing and regulate privacy through both (a) their own verbal consent and (b) written consent from their authorized representative. Residents and authorized representatives negotiate privacy by deciding which information may be shared publicly, although in many cases the authorized representative takes on an authoritative role and may not communicate decisions back to the resident. For example, AT1 explained how relatives are concerned about the visibility of their family member's identity and dementia, *"The family member was concerned [about the exhibit]. 'What if people that my dad knows sees his paintings and they will know he has dementia?'"* This tension requires all stakeholders to weigh the benefits of sharing certain information (e.g., resident's identity) against possible

negative outcomes (e.g., revealing a condition). To address this, some art therapists remove identifying information, “*I just still try to hide it [resident’s name], get rid of it and just put initials,*” (AT1). Here we see that vulnerability may compromise one’s voice (e.g., by removing identity) in an effort to guard against risks of sharing in the public region.

Deriving Benefits from a Co-present & Visible Audience

Sharing in art therapy may provide social and emotional support for residents, build a sense of community, and establish awareness of the abilities of people with dementia [4]. Yet, we observed that deriving these benefits across various regions of sharing depends on whether and how an audience is co-present and visible during the sharing experience. Art therapists emphasized the value of sharing artwork to scaffold conversation among a group of participants or in one-on-one therapy. AT1 said, “*Therapy is mainly socialization...they will use the art to interact with each other in that way.*” AT3 explained that art therapy is about “*building that community of support through the art.*” Sharing fosters social engagement and community building in the primary and secondary regions as sharing occurs face-to-face and directly involves the resident. Sharing in-person often leads residents to communicate deeper thoughts or emotions associated with their artwork. Furthermore, the primary and secondary regions focus on self-expression of residents, including taking time to formulate one’s thoughts and receive and react to comments from peers.

In contrast, sharing in the surrogate and public regions focuses on conveying an individual’s abilities to their social contacts, which may be more meaningful than sharing with one’s peers [39]. However, the resident may not be involved in sharing – they may not be physically present nor have visibility into who is viewing their work. In care plan meetings, residents do not often know who is viewing their artwork or how it is shared. Although this sharing focuses on conveying the abilities, strengths, and needs of a resident, the disinvolvement of the resident may magnify their vulnerability. Similarly, sharing in the public region helps establish awareness of abilities and communicates residents’ experiences, but this public sharing is against a backdrop of normative views of dementia as disease [17]. For example, AT1 noted how public exhibits help position older adults with dementia as contributors to society through their artwork, and said, “*People who live with dementia or Alzheimer’s disease, they are still capable of doing things that contribute to society.*” Primarily residents, family members, staff, and visitors attend exhibitions that occur within a care community. Exhibits outside of the care community create opportunities for exposure, advocacy, and participation with a broader audience but are less accessible to residents. In many cases, residents may not be present at or receive feedback during public exhibitions outside of the care community, affecting social or emotional benefits they derive from this form of sharing. However, when staff, visitors, or community members view the artwork alongside the resident, this serves to validate the residents’ abilities,

provide positive feedback, and potentially provide residents with a sense of control:

“*Because so many people [residents] are hearing like ‘no you can’t do that’ all the time and they are not allowed to make choices... So the fact that they were able to create something and then the same staff is saying like ‘wow you did a really good job, look at what you did’ I think that can make people feel like, ‘okay at least I have some control over something and look I did something positive,’*” (AT4).

Throughout our analysis, having a co-present and visible audience for sharing was important for residents’ personal enrichment. However, the categorization of these residents as vulnerable gives others license to share and speak on their behalf, at times isolating residents from potentially valuable interaction with their audience. Residents have few opportunities to interact with their audience in the surrogate and public regions, and we discuss below how online tools could help residents derive benefits from this interaction.

The Process and Product of Art Therapy

The literature debates the relative importance of the *process* of art therapy versus the *product* of art therapy [4]. In an effort to protect residents as a vulnerable population, our analysis finds that the focus shifts from sharing the process to sharing the product as one moves through each region. In this way, the material artifact or artwork created during art therapy serves as a boundary object [35], which is interpreted slightly differently in each region but retains a common structure that is understandable across these spaces. For example, sharing within the primary region focuses on sharing the act or in-the-moment experience of making art between a resident and therapist:

“*...when you are making art it’s just so focused in the moment and for [people with] Alzheimer’s that ‘in-the-momentness’ is really essentially what they have. I mean really essentially what all of us have but what they have more access to,*” (AT3).

Art making between a resident and therapist may focus on the kinesthetic experience of spreading paint on a canvas or manipulating fabric and intentionally avoid reflection on what is produced. Sharing within the secondary region also focuses on the in-the-moment process of making art but culminates with social interaction around the particular products of an art therapy session. That is, sharing may reflect some of what an individual was feeling during the process of making art, but the act of sharing is tied to the artifact the resident produced as a scaffold for conversation. And, sharing in the secondary region may focus on aesthetic qualities of the artifact to initiate conversation.

Similar to the secondary region, in the surrogate region, sharing is tightly coupled to both the process and product of art therapy. Art therapists rely on pieces of artwork when sharing during care plan meetings, but they use these artifacts to highlight aspects of the therapeutic process. For example,

notes written on the back of artwork capture residents' experiences while making art. AT4 explained:

"In the notes that I write about the session...I don't really write a lot [about] the artwork itself, I'm usually writing about the behaviors of the person..."

Art therapists use this documentation – both the artifact and their notes detailing behavior – in the surrogate region to convey the abilities and needs of the resident. Art therapists select pieces of artwork from the resident's portfolio when they provide an assessment to the authorized representative or clinical team, using both the process and product to convey the resident's abilities. AT2 said:

"I want to be able to explain to the family member the process... I think it can be misleading to just put it out there and let them come to their [own] conclusions."

In the public region, the focus of sharing is on the product and often highlights the artwork's aesthetic properties. Art therapists may curate public exhibitions to showcase visually appealing artwork that can be appreciated by a broader outside audience. At times, sharing in the public region may purposefully distance itself from the process of art therapy given the sensitive and private nature of making art and perceived risks of sharing about vulnerable individuals. There are occasions in which art therapists remove the individual's name or frame artwork without reference to the art therapy program, further distancing the product from the private and intimate context of therapy. Again, efforts to protect vulnerable individuals from risks in more public sharing may ultimately compromise their voices.

DISCUSSION

Our analysis of sharing within the context of art therapy for older adults with dementia contributes to CSCW in two primary ways. First, this work provides a case through which the field can think about sharing and privacy in the context of vulnerable populations. Second, we detail sharing practices in this particular context and from this discuss how new online systems could support regional sharing behaviors for older adults with dementia.

Understanding Vulnerability, Sharing, and Privacy

Older adults with dementia are a particular instance of a vulnerable population, and through this case we understand sharing as a much richer concept that encourages critical reflection on the ways in which online systems treat vulnerability. Information is not simply shared or unshared, but it involves a complex negotiation through multiple regions of public and private space [29]. Sharing in this context does not involve individualistic decision-making as embodied by mainstream content sharing platforms [13,26]. Rather, our analysis demonstrates how sharing is a social process, in which artifacts are co-owned by older adults and art therapists and privacy boundaries are cooperatively negotiated through moments of risk throughout these regions. That is, vulnerability is shaped by the context and information to be shared and the risks that therapists, authorized representatives, and others perceive. Sharing

online often involves setting public versus private permissions for information to be shared [8]. However, in this case artifacts mediate the social exchange of information about the self in ways that require navigating elements of risk, empowerment, and ownership, further shaping what constitutes vulnerability in sharing.

Our analysis makes salient several important questions for sharing and privacy involving vulnerable individuals. Is sharing always good? In this context, the value of sharing is in providing social or emotional support, building community, and promoting a sense of ability rather than disability. In this way, sharing is empowering and positive; yet, this is metered by the risks associated with disclosure of information during sharing (e.g., identity, emotional state, medical condition). How are vulnerable individuals empowered to share while managing the risks of revealing information? Towards this end, sharing may occur through surrogacy, which further highlights the ways in which sharing is a social process involving complex negotiations around deciding whether, what, and how to share. Prior work examines parents as surrogates in online information sharing about their children [3] but has yet to understand sharing involving surrogates at other points in life. Petronio and Altman [29] assert that when an individual is responsible for another person's information disclosure, errors in judgment and even deception may occur. This further highlights the importance of understanding sharing through surrogacy for vulnerable populations, who may be less able to detect such deception and rely on others to advocate on their behalf.

There is also a tension between policy and practice that further motivates this work and speaks to design for vulnerable populations more broadly. That is, while policies operating at multiple levels (e.g., HIPAA, residential rules) guide sharing behaviors and access to information, the ethics of when, how, and with whom sharing takes place are not well articulated in the field of clinical art therapy. Even for the practicing art therapist who is a co-author on this paper and those art therapists we interviewed, the ethics of what constitutes public versus private, how to gauge a resident's willingness to share (e.g., a head nod, verbal response, or dissenting glance), and what level of personal information should be represented through the artifact are blurred and open to interpretation. We believe this presents the greatest challenge to understanding how systems driven by algorithms and binary concepts of shared/un-shared and public/private information can support vulnerability.

Considerations for CSCW System Design

While our work details what sharing and privacy mean for older adults with dementia, we also revisit the social practices across various regions to inform the design of online systems to support this particular context.

Supporting Cooperative Negotiations of Privacy

Prior work notes that caregivers play an important role in regulating the boundaries of an older adult's possessions and territories [30]. In art therapy, artifacts created by older

adults with dementia are both co-owned and co-managed by others. As part of this, residents may not play an active role in sharing depending on the gradations of private versus public sharing. The resident's role in decision-making changes as more explicit privacy agreements (e.g., written consent) are needed in public sharing contexts, where there may be greater risk of revealing one's identity or condition. This introduces several important challenges for design – the resident's artwork may not be shared if their authorized representative does not agree; residents have no control over the public presentation of their work; they may not be present or receive feedback when social contacts view their work; and at times their identity as the creator may be concealed. While proxies often aid in communicating information about an older adult with dementia [28], online systems do not yet support the cooperative social practices and policies that guide information sharing in the surrogate and public regions. For example, future systems could offer mechanisms that allow the resident and their authorized representative to cooperatively specify the level and type of information disclosed based on the potential context or audience (e.g., groups of online social contacts). Further, these mechanisms could include information from the art therapist to clarify the circumstances of sharing, benefits of sharing, or attributes to share. More broadly, our work suggests a need for privacy settings that support cooperative decision-making and dynamic levels of involvement of others through moments of risk and vulnerability.

Increasing Audience Co-Presence and Visibility

Throughout our analysis we observed that the co-presence and visibility of one's audience is related to the benefits a resident derives from sharing. Data from art therapists suggest that an important area for design would be to support sharing with family members and close friends. The challenge, however, is related to the 'in-the-momentness' that older adults with dementia experience and asynchronous nature of many online systems. This requires new techniques for online sharing that give the perception that an audience is co-present or visible in a particular moment. For example, remote shared desktop experiences (e.g., IllumiShare [22]) could give the feeling of a co-present social contact during the art making experience. Asynchronous interaction, such as video messages, might allow a resident to revisit positive comments from social contacts about their artwork. Such design alternatives may even support the case of hypothetical sharing in the primary region, in which a video of a loved one could foster moments of sharing. While evocative, these systems would likely introduce new challenges to decision-making and regulating privacy.

Integrating the Process and Product into Social Sharing

Prior systems for art therapy focus on improving the therapist's activities [27] and neglect the opportunity to enhance the practice of making and sharing. Systems designed for this context should consider the ways in which artifacts can be shared without losing the richness and depth of information expressed through the artifacts. One way to

intertwine sharing the process and product is through the concept of digital artwork (e.g., [34]), in which everyday materials and objects are enhanced with digital traces to enable new forms of fluid online interaction. For example, Spyn embeds digital information into yard while knitting [34]. Embedding mechanisms for reciprocal communication with online contacts within art making tools or materials could support in-the-moment awareness of one's audience and foster sharing rich personal information through the interactive artifacts. This social exchange embedded in the artifact could potentially provide residents with further validation during moments of sharing, thus facilitating personal enrichment and wellbeing through art therapy.

CONCLUSION

Empowering older adults as active participants in creating and sharing online is of increasing interest to CSCW. Yet, important questions remain around how to design online systems to support the complex and dynamic social sharing practices of vulnerable populations. Art therapy for older adults with dementia is a rich context for examining these issues and re-envisioning the design of sharing systems. Our work highlights that online systems more broadly have yet to consider the complexities of vulnerability, sharing, and privacy, which raise important challenges for the field.

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REFERENCES

1. Irwin Altman. 1975. *The environment and social behavior: privacy, personal space, territory, crowding*. Brooks/Cole Pub. Co., Monterey, California, USA.
2. Alzheimer's Association. 2015. Alzheimer's Disease Facts and Figures. Retrieved May 7, 2015 from www.alz.org/facts/downloads/ff_quickfacts_2015.pdf
3. Tawfiq Ammari, Priya Kumar, Cliff Lampe, and Sarita Schoenebeck. 2015. Managing Children's Online Identities: How Parents Decide What to Disclose About Their Children Online. *Proc. of CHI '15*, ACM, 1895–1904. <http://doi.org/10.1145/2702123.2702325>
4. Renée L. Beard. 2012. Art therapies and dementia care: A systematic review. *Dementia* 11, 5: 633–656. <http://doi.org/10.1177/1471301211421090>
5. Scott Blunsden, Brandi Richards, Jen Boger, et al. 2009. Design and prototype of a device to engage cognitively disabled older adults in visual artwork. *In Proc. of PETRA '09*, ACM Press, 1–8. <http://doi.org/10.1145/1579114.1579162>
6. Mark Blythe, Peter Wright, John Bowers, et al. 2010. Age and experience: ludic engagement in a residential care setting. *In Proc. of DIS '10*, ACM Press, 161–170. <http://doi.org/10.1145/1858171.1858200>
7. A.J. Bernheim Brush, Kori M. Inkpen, and Kimberly

- Tee. 2008. SPARCS: exploring sharing suggestions to enhance family connectedness. *In Proc. of CSCW '08*, ACM Press, 629–638. <http://doi.org/10.1145/1460563.1460661>
8. Rajarshi Chakraborty, Claire Vishik, and H Raghav Rao. 2013. Privacy Preserving Actions of Older Adults on Social Media: Exploring the Behavior of Opting out of Information Sharing. *Decis. Support Syst.* 55, 4: 948–956. <http://doi.org/10.1016/j.dss.2013.01.004>
 9. Kathy Charmaz. 2006. *Constructing grounded theory: a practical guide through qualitative analysis*. SAGE Publications. <http://doi.org/10.1016/j.lisr.2007.11.003>
 10. Kathy Charmaz. 2008. Constructionism and the Grounded Theory Method. In *Handbook of Constructionist Research*, James A. Holstein and Jaber F. Gubrium (eds.). Guilford Press, 397–412.
 11. Raymundo Cornejo, Mónica Tentori, and Jesús Favela. 2013. Ambient Awareness to Strengthen the Family Social Network of Older Adults. *Computer Supported Cooperative Work* 22, 2-3: 309–344. <http://doi.org/10.1007/s10606-012-9166-2>
 12. Betsy van Dijk, Pavan Dadlani, Aart van Halteren, and Margit Biemans. 2010. Life changes, connection stays: photo sharing and social connectedness for people with special needs. *Proc. of ECCE '10*, ACM Press, 135–142. <http://doi.org/10.1145/1962300.1962327>
 13. Joan Morris DiMicco and David R Millen. Identity Management: Multiple Presentations of Self in Facebook. *Proc. of GROUP '07*, ACM, 383–386. <http://doi.org/10.1145/1316624.1316682>
 14. Mary Galvin. 2013. Who Is The Vulnerable One? *Proc. of CHI '13 Workshop on "Designing for-and with- Vulnerable People."*
 15. Peter Gregor, Alan F. Newell, and Mary Zajicek. 2002. Designing for dynamic diversity: interfaces for older people. *In Proc of Assets '02*, ACM Press, 151–156. <http://doi.org/10.1145/638249.638277>
 16. Greg Guest. 2006. How Many Interviews Are Enough?: An Experiment with Data Saturation and Variability. *Field Methods* 18, 1: 59–82. <http://doi.org/10.1177/1525822X05279903>
 17. Nancy Harding and Colin Palfrey. 1997. *The Social Construction of Dementia: Confused Professionals?* Jessica Kingsley.
 18. HIPPA Privacy Rule. 2007. 45 CFR. Parts 160, 164.
 19. Jesse Hoey, Kristis Zutis, Valerie Leuty, and Alex Mihailidis. 2010. A tool to promote prolonged engagement in art therapy: design and development from arts therapist requirements. *In Proc. of ASSETS '10*, ACM Press, 211–218. <http://doi.org/10.1145/1878803.1878841>
 20. Nancy Van House, Marc Davis, Morgan Ames, Megan Finn, and Vijay Viswanathan. 2005. The uses of personal networked digital imaging: an empirical study of cameraphone photos and sharing. *In Proc. of CHI '05*, ACM Press, 1853–1856. <http://doi.org/10.1145/1056808.1057039>
 21. Institute for Dementia Research & Prevention. 2012. Dementia. Retrieved May 1, 2015 from <http://idr.pbr.edu/faq.htm>
 22. Sasa Junuzovic, Kori Inkpen, Tom Blank, and Anoop Gupta. 2012. IllumiShare: sharing any surface. *In Proc. of CHI '12*, ACM Press, 1919–1928. <http://doi.org/10.1145/2207676.2208333>
 23. Da-jung Kim and Youn-kyung Lim. 2015. Dwelling Places in KakaoTalk: Understanding the Roles and Meanings of Chatrooms in Mobile Instant Messengers. *Proc. of CSCW '15*, ACM Press, 775–784. <http://doi.org/10.1145/2675133.2675198>
 24. Helen B. Landgarten. 1981. *Clinical Art Therapy: A Comprehensive Guide*. Taylor & Francis.
 25. Siân E Lindley, Richard Harper, and Abigail Sellen. 2008. Designing for elders: exploring the complexity of relationships in later life. *In Proc. of BCS-HCI '08*, British Computer Society, 77–86.
 26. Eden Litt, Erin Spottswood, Jeremy Birnholtz, Jeff T. Hancock, Madeline E. Smith, and Lindsay Reynolds. 2014. Awkward encounters of an “other” kind. *Proc. of CSCW '14*, ACM Press, 449–460. <http://doi.org/10.1145/2531602.2531646>
 27. Alex Mihailidis, Scott Blunsden, Jennifer Boger, et al. 2010. Towards the development of a technology for art therapy and dementia: Definition of needs and design constraints. *The Arts in Psychotherapy* 37, 4: 293–300. <http://doi.org/10.1016/j.aip.2010.05.004>
 28. Peter J. Neumann, Sally S. Araki, and Elane M. Gutterman. 2000. The Use of Proxy Respondents in Studies of Older Adults: Lessons, Challenges, and Opportunities. *Journal of the American Geriatrics Society* 48, 12: 1646–1654. <http://doi.org/10.1111/j.1532-5415.2000.tb03877.x>
 29. Sandra Petronio and Irwin Altman. 2002. *Boundaries of Privacy: Dialectics of Disclosure*. SUNY Press.
 30. Sandra Petronio and Samantha Kovach. 1997. Managing privacy boundaries: Health providers' perceptions of resident care in Scottish nursing homes. *Journal of Applied Communication Research* 25, 2: 115–131. <http://doi.org/10.1080/00909889709365470>
 31. Hayes Raffle, Cati Vaucelle, Ruibing Wang, and Hiroshi Ishii. 2007. Jabberstamp: embedding sound and voice in traditional drawings. *Proc. of IDC '07*, ACM Press, 137–144. <http://doi.org/10.1145/1297277.1297306>
 32. Yvonne Rogers and Gary Marsden. 2013. Does He Take Sugar?: Moving Beyond the Rhetoric of

- Compassion. *interactions* 20, 4: 48–57.
<http://doi.org/10.1145/2486227.2486238>
33. Yvonne Rogers, Jeni Paay, Margot Brereton, Kate L. Vaisutis, Gary Marsden, and Frank Vetere. 2014. Never too old: engaging retired people inventing the future with MaKey MaKey. *Proc. of CHI '14*, ACM Press, 3913–3922.
<http://doi.org/10.1145/2556288.2557184>
34. Daniela K. Rosner and Kimiko Ryokai. 2010. Spyn: augmenting the creative and communicative potential of craft. *Proc. of CHI '10*, ACM Press, 2407–2416.
<http://doi.org/10.1145/1753326.1753691>
35. Susan L. Star and James. R. Griesemer. 1989. Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39. *Social Studies of Science* 19, 3: 387–420.
<http://doi.org/10.1177/030631289019003001>
36. Ellen Greene Stewart. 2004. Art Therapy and Neuroscience Blend: Working with Patients Who Have Dementia. *Art Therapy* 21, 3: 148–155.
<http://doi.org/10.1080/07421656.2004.10129499>
37. John Vines, Róisín McNaney, Stephen Lindsay, Jayne Wallace, and John McCarthy. 2014. Special Topic: Designing for and with Vulnerable People. *interactions* 21, 1: 44–46. <http://doi.org/10.1145/2543490>
38. Jenny Waycott, Hilary Davis, Frank Vetere, et al. 2014. Captioned photographs in psychosocial aged care: relationship building and boundary work. *Proc. of CHI '14*, ACM Press, 4167–4176.
<http://doi.org/10.1145/2556288.2557290>
39. Jenny Waycott, Frank Vetere, Sonja Pedell, et al. 2013. Older adults as digital content producers. *Proc. of CHI '13*, ACM Press, 39–48.
<http://doi.org/10.1145/2470654.2470662>
40. Pamela Wisniewski, A.K.M. Najmul Islam, Bart P. Knijnenburg, and Sameer Patil. 2015. Give Social Network Users the Privacy They Want. *Proc. of CSCW '15*, ACM Press, 1427–1441.
<http://doi.org/10.1145/2675133.2675256>